Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Paper 4: Selection Functionals Reviews Response

Benjamin D. Summers <bsummers@adagiollc.com> Thu, Sep 17, 2026 at 12:15 AM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

---------- Forwarded message ---------
From: **ISMAIL-SUTTON, SARA R.** <sara.r.ismail-sutton@durham.ac.uk>
Date: Wed, Sep 16, 2026 at 10:28 AM
Subject: Re: Paper 4: Selection Functionals Reviews Response
To: Benjamin D. Summers <bsummers@adagiollc.com>

Hi Ben,

Thank you for the exceptionally detailed response. I have gone through your explanations, as promised.

The changes you describe appear to be directed appropriately at the main issues raised in my report, particularly the separation of uncertainty under the selected measure from uncertainty concerning the target, the withdrawal of the claimed implication from the kernel obstruction to selection bias, the distinction between the kernel and process settings, the revised calibration procedure, and the more limited positioning of Theorems A-C relative to standard SNIS theory.

As I have not reviewed the revised manuscript itself, I cannot confirm how fully or consistently these changes have been implemented in the text. However, I have only three remaining comments arising from the response itself.

First, the revised novelty claim is considerably more defensible. I would ensure that the novelty is attached specifically to the exact product identity, its finite-grid implementation, and the consequences derived from it in this setting. Multiplication of densities and the product-of-experts principle are themselves established ideas, so the fact that the construction multiplies rather than divides densities should not alone be presented as novel.

Second, one technical qualification may still be needed in the Mori–Zwanzig discussion. The one-sided matrix integral
[
\int_0^\infty \mathbb{E}[f(t)f(0)^{\mathsf T}],dt
]
is not in general itself symmetric and positive semidefinite merely by Bochner's theorem. The white-noise covariance intensity is normally obtained from the two-sided integrated autocovariance, or equivalently an appropriately symmetrized one-sided integral,
[
\int_0^\infty \left(R_f(t)+R_f(t)^{\mathsf T}\right),dt.
]
If the Mori assumptions reduce this to \gamma C+C\gamma^{\mathsf T}, that symmetric quantity should be identified directly as the covariance intensity.

Third, I would qualify the statement that "the estimator has no limit." The selected measures and their estimands may have a continuum limit while the fixed-proposal importance-sampling representation becomes singular and its chi-square divergence grows without bound. That establishes failure of a stable fixed-proposal representation under refinement, but it does not by itself rule out every possible joint limit in which N grows with M, or an alternative sequence of proposals is used. Saying that the present importance-sampling representation has no nondegenerate fixed-proposal continuum limit would be more precise.

Apart from those qualifications, the explanations describe substantial and thoughtful revisions addressing the substance of my review. Confirming the final mathematical wording, cross-section consistency, and implementation throughout the revised manuscript would require a separate review of that version.

Best wishes,

Dr Sara Ismail-Sutton

Dr Sara Ismail-Sutton

From: Benjamin D. Summers <bsummers@adagiollc.com>
Sent: 16 September 2026 04:08

To: ISMAIL-SUTTON, SARA R. <sara.r.ismail-sutton@durham.ac.uk>; Ezequiel Boero <ezequiel.boero@adagioinstitute.org>
Subject: Paper 4: Selection Functionals Reviews Response

[EXTERNAL EMAIL]

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